Appendix 2 solutions

1

Addition $A + B$ is undefined because dimensions differ. The products AB and BA have sizes 2×2 and 3×3, respectively. Transpose A^T is 3×2, so $A^T C$ exists and is 3×2.

2

$$AB = \begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix}, \qquad BA = \begin{pmatrix} 3 & 4 \\ 1 & 2 \end{pmatrix},$$

which differ. Also

$$B + C = \begin{pmatrix} 1 & 0 \\ 3 & 0 \end{pmatrix}, \qquad A(B + C) = \begin{pmatrix} 7 & 0 \\ 15 & 0 \end{pmatrix}.$$

Directly,

$$AB + AC = \begin{pmatrix} 2 & 1 \\ 4 & 3 \end{pmatrix} + \begin{pmatrix} 5 & -1 \\ 11 & -3 \end{pmatrix} = \begin{pmatrix} 7 & 0 \\ 15 & 0 \end{pmatrix},$$

confirming distributivity.

3

Expanding the first row,

$$\det A = 2 \det \begin{pmatrix} 3 & 1 \\ -1 & 2 \end{pmatrix} - \det \begin{pmatrix} 0 & 1 \\ 1 & 2 \end{pmatrix} + 0 = 2(6 + 1) - (0 - 1) = 15.$$

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4

The determinant is $4 \cdot 2 - 1 \cdot 7 = 1 \neq 0$, so

$$A^{-1} = \begin{pmatrix} 2 & -1 \\ -7 & 4 \end{pmatrix}.$$

Verification gives

$$\begin{pmatrix} 2 & -1 \\ -7 & 4 \end{pmatrix} \begin{pmatrix} 4 & 1 \\ 7 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

5

For arbitrary i, j,

$$((AB)^T)_{ij} = \sum_k A_{jk} B_{ki} = \sum_k (B^T)_{ik} (A^T)_{kj} = (B^T A^T)_{ij}.$$

The dummy index k ranges over the inner dimension, and equality of all entries proves the result.

Take $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$. Both determinants are one, but

$$AB = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \neq \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} = BA.$$

6

The coefficient matrix $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$ has determinant five and inverse $\frac{1}{5}\begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix}$. Thus

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{5}\begin{pmatrix} 15-7 \\ -5+14 \end{pmatrix} = \begin{pmatrix} 8/5 \\ 9/5 \end{pmatrix}.$$

Cramer gives the same values: $x = (15-7)/5$, $y = (14-5)/5$.

After changing the first equation,

$$A' = \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix}, \qquad b' = \begin{pmatrix} 10 \\ 7 \end{pmatrix}, \qquad \det A' = 10 \neq 0.$$

Scaling an entire equation preserves its solutions, so the unique pair stays $(8/5, 9/5)$. One must use the new inverse, not multiply the old inverse by the new right side:

$$(A')^{-1} = \frac{1}{10}\begin{pmatrix} 3 & -2 \\ -1 & 4 \end{pmatrix}, \qquad (A')^{-1}b' = \begin{pmatrix} 8/5 \\ 9/5 \end{pmatrix}.$$

This example remains nonsingular. No-solution versus infinitely-many branches are needed only when the new system is actually singular.

7

Apply $R_2 - 2R_1$. For $b = (3, 6)^T$, the second augmented row is $[0, 0|0]$. Set $y = t$, then $x = 3 - 2t$, with all real t. For $b = (3, 7)^T$, it is $[0, 0|1]$, the contradiction $0 = 1$. A zero determinant rules out an inverse but says nothing about whether b belongs to the column space.

8

The entry $(A \operatorname{adj} A)_{ij} = \sum_k a_{ik}C_{jk}$ equals $\det A$ when $i = j$. Otherwise it is the expansion after replacing row j by row i, producing repeated rows and zero. Thus $A \operatorname{adj} A = (\det A)I$; the column argument gives the other identity. Divide by nonzero determinant to obtain an inverse. Conversely, an inverse gives $\det A \det(A^{-1}) = 1$, proving nonvanishing.

For $A = B = I_2$, $\det(A + B) = 4$, while $\det A + \det B = 2$. Scaling the whole matrix scales each of n rows by c, so multilinearity gives c^n, rather than a single factor c.

9

$$BA = (1), \qquad AB = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \neq I_2.$$

Here B is a left inverse of A, and A a right inverse of B. Applying B then A to a two-dimensional vector erases its second coordinate. The square-matrix theorem that a one-sided inverse is two-sided relies on matching finite dimensions and cannot cross these dimensions.

10

If integer matrices satisfy $AB = BA = I$, their integer determinants multiply to one, forcing $\det A = \pm 1$. Conversely, integer minors make $\operatorname{adj} A$ integer-entry; dividing by ± 1 gives an integer two-sided inverse.

Determinant two still permits a real inverse. For example, $\operatorname{diag}(2, 1, \ldots, 1)$ has inverse $\operatorname{diag}(1/2, 1, \ldots, 1)$, which is not integer-entry. Real invertibility and being a unit in the integer matrix ring depend on different allowed entries.

11

Direct multiplication gives $N^2 = M^2 = 0$, $NM = \operatorname{diag}(1, 0)$, and $MN = \operatorname{diag}(0, 1)$. Thus $(N + M)^2 = NM + MN = I_2$, whereas $N^2 + 2NM + M^2 = \operatorname{diag}(2, 0)$. The general expansion is $A^2 + AB + BA + B^2$; when $AB = BA$, its middle terms combine to $2AB$.

References and attribution

Lin Cing, handwritten linear algebra notes. Ralph P. Grimaldi, Discrete and Combinatorial Mathematics, 5th ed., Appendix 2.

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